

Integrating mechanistic data to contextualize in vivo apical observations using the developmental neurotoxicity in vitro battery

Bianna L. Jackson ¹, Yaoxing Wu¹, Kian Afsharian ², Christopher Schlosser¹, Angela H. Hofstra², Laura M. Minnema¹, Jödis Klose ³, Ellen Fritsche ^{3,4}, Manasvinee Mayil Vahanan^{5,6}, Timothy J. Shafer ⁶, Kelly E. Carstens⁶, Agnes L. Karmaus ^{1,7,*}

¹Syngenta Crop Protection LLC, North America Human Safety, Greensboro, NC, United States

²Syngenta Canada, Inc, North America Human Safety, Guelph, ON, Canada

³DNTOX GmbH, Düsseldorf, Germany

⁴Swiss Centre for Applied Human Toxicology, Basel, Switzerland

⁵Oak Ridge Institute for Science and Education (ORISE), Oak Ridge, TN, United States

⁶Center for Computational Toxicology and Exposure, Office of Research and Development, United States Environmental Protection Agency, Research Triangle Park, NC, United States

⁷Present address: Sciome, LLC, Research Triangle Park, NC, United States

*Corresponding author: Syngenta Crop Protection, LLC, 410 South Swing Road, Greensboro, NC 27409, United States. Email: agnes.karmaus@sciome.com

Abstract

Regulatory toxicity studies traditionally rely on in vivo apical endpoints for hazard characterization and risk assessment; when available, relevant and reliable in vitro assays are anticipated to aid this interpretation. Developmental neurotoxicity (DNT)-related parameters such as brain morphometry can be especially challenging to interpret, based on technical challenges such as sample preparation artifacts. This study employed the Developmental Neurotoxicity In Vitro Battery (DNT-IVB), designed to provide functional insights on key neurodevelopmental processes (KNDPs), to provide context for equivocal brain morphometry effects observed in a legacy in vivo DNT study conducted with acibenzolar-S-methyl (ASM), a plant host defense inducer. ASM and its major mammalian metabolite, acibenzolar acid (AA), were tested in all 17 in vitro assays comprised within the current iteration of the DNT-IVB. In vitro testing concentrations were derived to match the existing in vivo DNT study by applying physiologically based pharmacokinetic (PBPK) modeling with adjustment for rat versus human toxicokinetics. All concentration-response DNT-IVB assay data were analyzed using the US EPA ToxCast Pipeline. ASM showed no effect on any KNDP in vitro. AA was active only in the rat network formation assay with low potency AC50 ranges of 96.1 to 150 μ M that modeling shows will not be achieved in vivo at doses where the apical effects in question were observed. This case study demonstrates the potential of the DNT-IVB to be used as a tool to assess equivocal findings from in vivo DNT studies and contribute to a weight of evidence to support conclusions on chemical-mediated DNT.

Keywords: developmental neurotoxicity; new approach methodologies; in vitro; physiologically based pharmacokinetic modeling

The evaluation of developmental neurotoxicity (DNT) for environmental chemicals has historically consisted of in vivo test protocols, i.e. OECD TG 426, OPPTS 870.6300 (US EPA 1998; OECD 2007). These tests involve exposing pregnant rats to a test chemical from implantation through lactation followed by assessment in offspring of growth, neurodevelopmental landmarks, sensory and motor behaviors, cognitive function, and neuropathology including brain morphometry. The types of measurements can be difficult to conduct and the abundance of resulting data can be difficult to interpret (Li et al. 2017).

Among the many evaluated endpoints in the in vivo DNT study, measurements of brain regions are collected during postnatal development and examined for evidence of neuropathological alterations. Treatment-related reduction in the size of brain region(s) is assumed to be indicative of perturbed

neurodevelopment, possibly through a reduction in cell number and/or processes (OECD 2007). Within this context, brain morphometry refers to linear or areal quantitative measurements of regions within the brain that are often anchored to a neuroanatomical landmark. Effects are examined along with all other study endpoints in a weight of evidence (WoE) to derive conclusions on potential DNT (Garman et al. 2016). A high level of expertise and rigorous standardization of techniques is required to produce homologous histological sections to make meaningful comparisons across dose groups. Prior to the updated OECD guidance in 2007, tissues from the control and highest dose group were examined first; if an effect of treatment was observed, examination of tissues from lower dose groups was subsequently made. This stepwise approach for evaluating brain morphometry has been highly scrutinized as a liability due to artifactual batch

inconsistencies introduced from the extended time for some samples to be stored in fixative (Bolon et al. 2006; Tsuji and Crofton 2012; Garman et al. 2016; Li et al. 2017). Thus, the updated (current) OECD guidance recommends that, “brain tissues collected for morphometric analysis should be embedded in appropriate media at all dose levels at the same time in order to avoid shrinkage artifacts that may be associated with prolonged storage in fixative” (OECD 2007).

Herein we present an previously unpublished in vivo DNT study conducted according to former OECD guidelines for acibenzolar-S-methyl (ASM), a plant host defense inducer. Tissues for morphometric analysis were stored in fixative for varying lengths of time making it unclear if decreases in morphometric measurements in the rat dorsal cortex and cerebellum observed in male rats from the low- and mid-dose groups on postnatal day (PND) 63 were compound-related or technical artifacts. Increased auditory startle response (ASR) in females at the highest dose group on PND 23 was also observed. Altogether, these findings were difficult to interpret and resulted in varying regulatory implications (PMRA 2011; EFSA 2014; US EPA 2017) identifying a need to further investigate and contextualize these results.

Fit-for-purpose in vitro assays have been developed into a battery endorsed by the OECD (OECD 2023) to yield insight into several processes of nervous system development at the cellular level. This battery, termed the Developmental Neurotoxicity In Vitro Battery (DNT-IVB), comprised 17 in vitro assays which model and assess chemical effects on key neurodevelopmental processes (KNDPs) and can detect the majority of chemicals considered to be potential in vivo developmental neurotoxicants (Harrill et al. 2011; Shafer 2019; Masjosthusmann et al. 2020). Therefore, in vitro assessment of chemical-mediated effects on KNDPs utilizing the DNT-IVB may be used to assess if putative apical findings in an in vivo DNT study are likely to be compound related.

This case study leverages the DNT-IVB to provide insights on KNDPs that may underlie the chemical-related findings observed in the in vivo DNT study conducted with ASM in rats. Both ASM and its primary mammalian metabolite, acibenzolar acid (AA), were tested in all 17 DNT-IVB assays and data were analyzed using the US EPA ToxCast pipeline for consistency with prior DNT IVB data. In silico physiologically based pharmacokinetic (PBPK) modeling was applied to ensure observed activity in the in vitro assays could be compared against in vivo observations. This case study demonstrates the utility of the DNT-IVB as a complement to an in vivo DNT study in understanding chemical-mediated effects in a fit-for-purpose manner.

Materials and methods

Chemicals

The test articles used for this case study were the pesticide ASM (CASRN 135158-54-2; DTXSID 1032519; 99.0% purity) and its primary mammalian metabolite, AA (97.6% purity), which have the common names of S-methyl benzo[1,2,3]thiadiazole-7-carbothioate (CASRN 135158-54-2; DTXSID1032519) and benzo [1,2,3]thiadiazole-7-carboxylic acid (CASRN 35272-27-6; DTXSID40885616), respectively. With AA representing up to 92% of the administered dose among urinary metabolites detected in rats, it was deemed to be the only relevant mammalian metabolite included in this study. Both chemicals were supplied by Syngenta Crop Protection, LLC (Greensboro, NC, United States). Testing facilities prepared stock solutions in DMSO and titrated

to final concentration series (6 to 9 concentrations depending on the assay) with a final DMSO concentration of 0.1% in media.

The testing concentrations in the DNT-IVB assays were chosen to be biologically and toxicologically relevant to inform on the in vivo DNT findings of concern. Testing concentrations were selected based on the highest dose level tested in the in vivo DNT study with ASM. To ensure testing concentrations encompass the doses of the in vivo DNT study, PBPK modeling was used to derive equivalent internal concentrations expected in rat and human fetal blood and brain which were used as testing concentrations in vitro.

In vivo DNT study

A previously conducted in vivo guideline DNT study for ASM was retrieved and evaluated. The study was conducted according to OPPTS 870.6300 (US EPA 1998). Briefly, ASM (purity 97.9%) was administered to groups of 30 time-mated female Alpk: AP₁SD (Wistar-derived) rats in diet containing 0, 100, 1,000, or 4,000 ppm ASM (corresponding to 0, 8.2, 82, and 326.2 mg/kg bw/day; dietary concentrations were consistent throughout the study, e.g. were not adjusted during late gestation or lactation). The pregnant rats were 10 to 12 wk old and weighed 220 to 300 g on delivery. CT1 diet (Special Diets Services Limited, Witham, Essex, UK) was the control substance and vehicle. The female rats were housed individually in solid plastic cages with sawdust bedding and environmental enrichment (“nylabones” and “fun tunnels”). Test diets were provided continuously from gestational day (GD) 7 to PND 22. The day of sperm-positive vaginal smear was designated as GD 1. Pups were allocated to the F1 phase of the study on PND 5, separated from the dam on PND 29, housed separately (i.e. up to 4 males or 4 females per cage), and allowed to grow to adulthood. The animal room was designed to give 12 h light: Dark at 22±3°C and 30% to 70% relative humidity. Evaluations in pups included observations on clinical condition and body weight (on PND 5, 12, 18, 22, 29, 36, 43, 50, 57, and 63), developmental landmarks (daily from PND 29 (females) and 41 (males)), a functional observation battery (FOB) (on PND 5, 12, 18, 22, 36, 46, and 60), effects on motor activity (on PND 14, 18, 22, and 60), response to auditory startle (on PND 23 and 61), assessment of learning and memory (on PND 21/24 and 59/62), and neuropathological evaluation, including linear morphometry and brain weights (on PND 12 and 63). For more details on the endpoints examined and statistical methods employed, refer to File S1.

PBPK modeling

To establish relevant testing concentrations for in vitro testing, a multi-step PBPK modeling approach was used to predict fetal exposure. This process integrated rat and human nonpregnant and pregnant/gestational modeling (Fig. 1). It is important to note that ASM is rapidly and extensively metabolized such that the major acid metabolite (AA) is predicted to have greater internal exposures. Thus, our approach for in vitro testing was to evaluate both of these chemicals in isolation.

First, nonpregnant rat PBPK models for the parent compound ASM and primary metabolite AA were developed. PBPK models for ASM and AA were developed in accordance with the World Health Organization (WHO 2010) and US EPA (2006) guidelines. In vitro absorption, distribution, metabolism, and excretion (ADME) studies using rat and human tissues were employed to generate chemical-specific input parameters for ASM and AA. In silico parameters were obtained using the Schmitt (2008) algorithm within the High-Throughput Toxicokinetics (httk) R package (Pearce et al. 2017). The nonpregnant rat model was evaluated

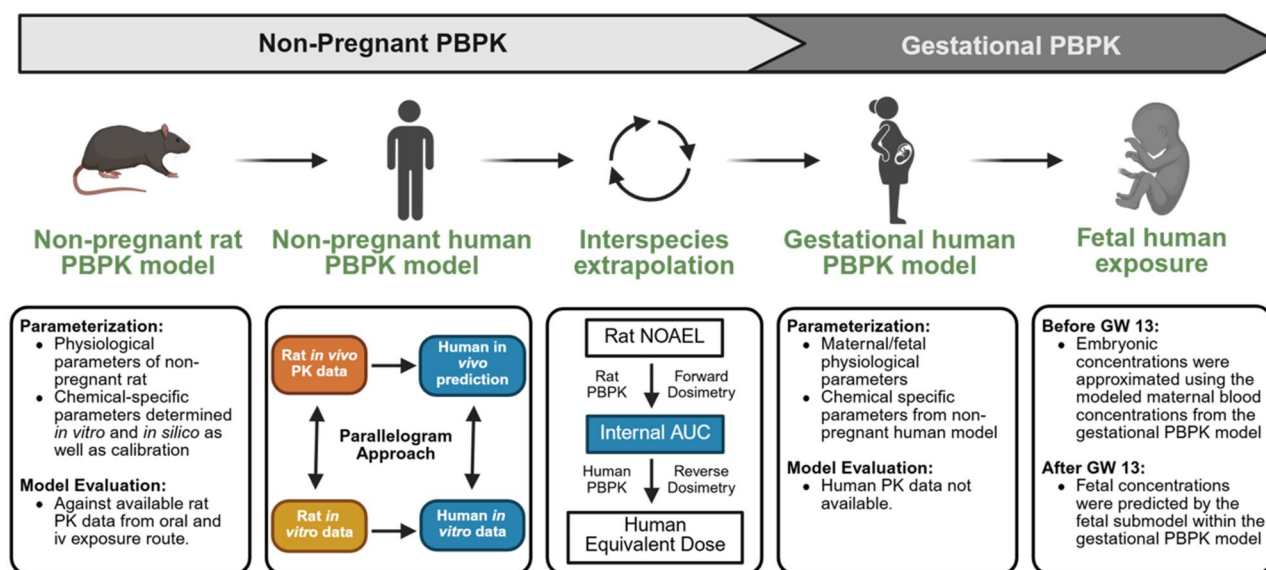


Fig. 1. Physiologically based pharmacokinetic (PBPK) modeling workflow. A nonpregnant rat PBPK model was developed and parameterized using *in vitro* and *in silico* data. The rat model was evaluated using rat *in vivo* pharmacokinetic (PK) results from oral and intravenous (iv) routes. A parallelogram approach was employed to conduct specific extrapolation between rat and human to develop the nonpregnant human PBPK model. The nonpregnant human model was extended to a gestational human PBPK model by incorporating physiological changes during pregnancy. This model was used to predict the internal exposure of fetus to ASM and AA at different gestational ages (specifically before and after gestational week (GW) 13), given a daily external dose level equivalent to the human equivalent dose (HED) at the *in vivo* low dose (rat no observed adverse effect level (NOAEL): 100 ppm; 8.2 mg/kg bw/d, corresponding to an HED of 1.51 mg/kg bw/d) or the level equivalent to the highest dose tested *in vivo* (4,000 ppm; 326.2 mg/kg bw/d, corresponding to an HED of 62.7 mg/kg bw/d). The resulting maximum daily average blood and brain concentrations were used to select the nominal concentration ranges in the DNT-IVB assays. AUC, area under the curve.

using rat *in vivo* pharmacokinetic data. Second, rat to human species extrapolation was conducted using a parallelogram approach to develop the nonpregnant human PBPK model (US EPA 2007; Motwani et al. 2017; Kuepfer et al. 2023). The parallelogram approach was used as it optimally leverages the relationship between the *in vitro* and *in vivo* findings in the rat and the relationship between *in vitro* rat and *in vitro* human findings. Third, the nonpregnant rat and human PBPK models were used to derive human equivalent doses (HEDs). Fourth, the nonpregnant human model was extended to pregnancy by incorporating physiological changes in pregnancy to develop the gestational human PBPK model (Kapraun et al. 2022). Of note, we conservatively assumed that with perfusion-limited placenta, the redistribution of the chemicals occurs instantaneously between the maternal and fetal placenta tissues at any given moment (Szeto et al. 2021). Therefore, the transfer rate across the placenta is assumed to be rate-determined by maternal or fetal placental blood flow (Kapraun et al. 2022), which might result in overestimation of fetal exposure. Last, the gestational human model was used to predict the fetal HED level equivalent to the highest dose tested *in vivo*. The modeled maximum 24-h (daily) average concentrations were used to guide representative testing concentration selection for DNT-IVB testing.

A summary of the model structures, physiological and chemical-specific parameters for the PBPK models, and additional details pertaining to PBPK modeling are presented in File S2. Briefly, the maximum 24-h average concentration was selected over the peak concentration (C_{max}) as the internal dose metric for *in vitro* to *in vivo* extrapolation (EFSA et al. 2025; Lin et al. 2025). DNT is driven by sustained disruption of neurodevelopmental processes rather than transient peak exposures, making an integrated measure of exposure more biologically relevant. This also more accurately mimics DNT-IVB exposures which employ long-duration, constant-concentration exposures

(days to weeks) that are more analogous to *in vivo* average concentrations than to a transient C_{max} spikes. Because gestation is an inherently dynamic process during which a true PK steady state is not reached, the maximum of the 24-h (daily) average blood and brain concentrations across the simulated gestational period was used, capturing the window of highest sustained fetal exposure.

Simulations, statistical analysis, and data visualization were carried out using R v4.5.0 (R Core Team 2024). Differential equations were solved using *deSolve* v1.40 (Soetaert et al. 2010). Monte Carlo simulations were carried out to characterize uncertainty and biological variability in exposure under defined dosing scenarios. Details are provided in File S2.

Assays in the DNT-IVB

DNT-IVB testing was performed as described in the OECD-endorsed testing procedure (OECD 2023). Brief summaries of all assay systems are provided below. For any additional details pertaining to test method conduct, please refer to the OECD endorsement (OECD 2023) or primary literature describing the development of these test systems (as cited within the respective section below).

Proliferation: NPC1 assay

Primary human NPCs (hNPCs) were derived from fetal cortex (Lonza, Verviers, Belgium; gestational weeks 16 to 19) and cultivated as free-floating 3D neurospheres in proliferation medium. Neurospheres were cultured under standard conditions at 37°C and 5% CO₂ in 10cm diameter cell culture dishes coated with poly-2-hydroxyethyl methacrylate (poly-Hema; Merck; United States). Once a week, hNPCs were passed mechanically into cubicles of 0.2 mm edge length using a tissue chopper and in order to refresh the culture medium, half of the medium was replaced thrice a week. For hNPC proliferation assessment, 2 to 3 days

after passaging, 0.3-mm-sized neurospheres were plated on poly-Hema-coated 96-well U-bottom plates (one neurosphere per well) in 100 μ l proliferation medium and exposed to 7 concentrations of each test compound with 4 replicates per concentration. hNPC proliferation was validated after 72 h of chemical exposure by assessing the incorporation of bromodeoxyuridine (BrdU, Roche, Switzerland) during DNA synthesis via a luminescence-based BrdU ELISA assay kit (Roche, Switzerland). Cytotoxicity and cell viability were assessed in parallel at the end of the exposure, by using a membrane integrity assay (CytoTox-ONE assay, Promega) measuring the lactate dehydrogenase (LDH) release into the supernatant, and by performing an Alamar Blue assay (Cell Titer Blue, Promega), respectively. Each plate included a positive control for proliferation arrest (proliferation medium without growth factors), BrdU background control, lysis control, and background control, as described by the developers. For hNPC proliferation experiments, at least 3 independent biological replicates were performed. Experiments were defined as independent if they were generated with hNPCs from different individuals or a different passage number. For complete details, see Masjosthusmann et al. (2020) and Koch et al. (2022).

Proliferation: HNP1 Prolif assay

hNPCs were derived from a neuroepithelial cell lineage of WA09 embryonic stem cells (ArunA Biomedical, Athens, GA) and cultivated as a 2D monolayer. Cells were plated at 1.5×10^4 cells/well on 96-well plates with proliferation medium. Forty-eight hours after plating, cells were treated with compounds (7 concentrations/compound) and returned to the incubator. Triplicate plates were prepared to yield a total of 3 technical replicates. Each plate included 8 wells each of 10 μ M aphidicolin as a positive control for inhibition of proliferation and the vehicle control. Proliferation was assessed by treating cells with BrdU 20 h after chemical exposure and fixing them 4 h later. Immunocytochemical (ICC) staining of BrdU-positive cells was used to quantify proliferation by quantitative fluorescence imaging was performed with Cellomics ArrayScan VTI HCS Reader (Thermo Fisher Scientific). Cell viability was determined in sister plates using the CellTiter-Glo 2.0 luminescent assay (Promega) which measures the amount of ATP present in each well. Cells were seeded in white opaque 96-well plates precoated with poly-L-ornithine and laminin. Two hours after plating, cells were exposed to chemicals and returned to the incubator. Twenty-four hours after chemical treatment, 50 μ l of CellTiter-Glo 2.0 Reagent was added to each well and the plate incubated for 30 min at room temperature. Luminescence was then recorded in each well using a FLUOstar Optima plate reader (BMG LABTECH, Cary, NC). Each plate included 8 wells with staurosporine (0.1 μ M) as a positive control for apoptosis and the vehicle control. The experiment was repeated across 3 plates. For complete details, see Harrill et al. (2018).

Apoptosis: HNP1 Apop assay

NPCs derived from human ESC WA09 line were plated on opaque 96-well plates. Two hours after plating, cells were exposed to 10 concentrations of each test compound in triplicate. After 24 h, apoptosis was measured by detection of activated caspase-3/7 (Caspase-Glo 3/7 assay kit, Promega) (Druwe et al. 2015; Harrill et al. 2018). Each plate included 8 wells with staurosporine (0.1 μ M) as a positive control for apoptosis and the vehicle control. The experiment was repeated across 3 plates.

Migration: UKN2 (cMINC) assay

Neural crest cells (NCCs) were derived from induced pluripotent stem cells (iPSCs) according to Nyffeler et al. (2017). Cells were seeded on 96-well plates around stoppers at a density of 20,000 cells/well. After the removal of the stopper, they were allowed to migrate into the cell-free area. Cells were exposed to 6 concentrations of each chemical in triplicate for 24 h, then stained with calcein-AM (viable cells) and Hoechst-33342 (nuclei). Wells were imaged on a Cellomics ArrayScan VTI high-content imaging microscope (Cellomics, PA). The numbers of migrated double-positive cells were quantified by the RingAssay software (<http://invitrotox.uni-konstanz.de>). Cell viability was assessed using an automated image analysis algorithm (ArrayScan VTI 700 Series software, v. 7.6.2.4), defining double-positive cells as viable. Each plate included positive controls for migration arrest (0.2 μ M cytochalasin D) and the vehicle control. The experiment was repeated 3 times. A detailed description is found in Krebs et al. (2020).

Migration: NPC2 assay

hNPC migration and differentiation assays (NPC2-5) are multiplexed. By plating 3D neurospheres (0.3 mm) into 100 μ l differentiation medium on poly-D-lysine (PDL) and Laminin-coated 96-well plate, neurospheres settle down and hNPCs radially migrate out of the sphere core and differentiate into radial glia, neurons, and oligodendrocytes. A detailed description of the NPC2-5 assays can be found in Masjosthusmann et al. (2020) and Koch et al. (2022). In brief, neurospheres were exposed up to 120 h to 7 concentrations of each test compound with 4 replicates per concentration. For NPC2-5 experiments, at least 3 independent biological replicates were performed. Experiments were defined as independent if they were generated with hNPCs from different individuals or a different passage number. After 72 h of exposure, radial glia migration (NPC2a) was assessed using brightfield microscopy (TECAN Spark Cyto) of unstained cells by measuring the size of the scaffold constructed by the migrated radial glia (full migration area). After 72 h, the medium including compounds was replenished, and cells were allowed to further migrate and differentiate for additional 48 h, after which cells were fixed with 4% PFA and immunocytochemically stained for TUBB3 (neurons), O4 (oligodendrocytes), and Hoechst-33258 (nuclei). High content image analyses (HCA) and an convolutional neural networks (CNN)-based in-house developed software "CytoPixel" were used to automatically acquire information on radial glia migration (NPC2a) based on the Hoechst-stained cell nuclei within the migration area of the differentiated spheres, and to identify immunostained neurons (TUBB3) and oligodendrocytes (O4) to calculate radial glia cell (NPC2a), early neuronal (NPC2b), and oligodendrocyte (NPC2c) migration (Schmuck et al. 2017; Masjosthusmann et al. 2020). Thereby, NPC2b and NPC2c were defined as mean distance of all neurons/oligodendrocytes within the migrations area divided by radial glia migration distance. As an endpoint-specific positive control, 0.1 mM src kinase inhibitor 3-(4-chlorophenyl) 1-(1,1-dimethylethyl)-1H-pyrazolo [3,4-d]pyrimidin-4-amine (PP2) was used for radial glia migration arrest (Baumann et al. 2015; Nimtz et al. 2019). Cytotoxicity was assessed by CytoTox-ONE assay after 72 and 120 h and cell viability was assessed by Alamar Blue assay after 120 h. Each plate included positive controls for radial glia migration arrest as well as lysis control, background control, and vehicle control.

Differentiation: NPC3 and NPC5 assays

During hNPC migration out of the plated neurosphere (NPC2), cells differentiate into radial glia, and further into β -III-tubulin (TUBB3)-positive neurons and O4-positive oligodendrocytes over the period of 120 h (Nimtz et al. 2019; Masjosthusmann et al. 2020; Koch et al. 2022). Neurons and oligodendrocytes were automatically annotated based on ICC staining using a machine learning software based on 2 CNNs (Förster et al. 2022) and an in-house developed image analysis software “CytoPixel.” Thereby, the number of annotated β -III-tubulin and O4-positive cells was normalized to the total number of Hoechst-positive nuclei. Positive controls for neuronal differentiation inhibition (1 ng/ml LEFG) and oligodendrocyte differentiation inhibition (2 ng/mL BMP2) (Masjosthusmann et al. 2018) were included on each plate.

Neurite outgrowth: NPC4 assay

The NPC4 assay is an extension of the NPC3 assay (neuronal differentiation). Following 120 h of exposure, neurite area (NPC4a) and neurite length (NPC4b) were defined as the mean area in pixels (without nuclei) and mean length in μ m, respectively, of all β -III-tubulin-positive neurons identified by the CNN and a skeletonization algorithm using the in-house developed software “CytoPixel” (Schmuck et al. 2017; Förster et al. 2022). Each plate included a positive control for impaired neurite outgrowth (0.01 μ M narciclasine) and a vehicle control.

Neurite outgrowth: UKN4 (NeuriTox) assay

Human neural stem cells (NSCs) derived from the mesencephalon of an 8-wk-old fetus (LUHMES) were differentiated into morphologically and biochemically mature dopamine-like neurons as described previously (Lotharius et al. 2005; Scholz et al. 2011; Krug et al. 2013; Gutbier et al. 2018). Proliferating cells were seeded on 96-well plates in differentiation medium and allowed to differentiate for 48 h, after which cells were exposed to 9 concentrations of each test chemical in triplicate for 24 h. Cells were stained with 1 μ g/ml H-33342 AND 1 μ M calcein-AM. Fluorescent imaging was performed (AxioObserver, Zeiss, Germany), then neurite area and viability were assessed as described previously (Stiegler et al. 2011). Image acquisition was performed using an Array-Scan VTI HCS Reader (Cellomics, PA, United States) equipped with a Hamamatsu ORCA-ER camera. Ten fields per well were imaged with 2 channels at 20x magnification (2x2 pixel binning). Excitation/emission wavelengths of 365 $\pm$ 50/535 $\pm$ 45 nm were used for Hoechst detection in channel 1 and 474 $\pm$ 40/535 $\pm$ 45 nm were used for calcein detection in channel 2. Nuclei were identified in channel 1 depending on intensity, area, size, and shape. Their outlines were expanded by 3.2 μ m to define a virtual cell soma area (VCSA). All calcein-positive pixels were defined as viable cellular structures (VCSs). The neurite area was automatically calculated by excluding the VCSAs from the VCSs. Morphologically normal nuclei co-localizing with VCSs were defined as live cells. Condensed nuclei or nuclei not overlaid with a calcein stain (VCSA) were considered dead. Each plate included a positive control for neurite outgrowth inhibition (0.05 μ M narciclasine) and the vehicle control (Krebs et al. 2020). The experiment was repeated independently 3 times.

Neurite outgrowth: UKN5 (PeriTox) assay

Immature dorsal root ganglia neurons differentiated from hiPSC (Hoelting et al. 2016; Holzer et al. 2022) were seeded on 96-well plates in differentiation medium. After 1 h, cells were exposed to 6 concentrations of each test chemical in triplicate for 24 h.

Endpoint measurements were exactly as for the UKN4 assay. Each plate included positive controls for neurite outgrowth inhibition (0.05 μ M narciclasine) and the vehicle control. The experiment was repeated independently 3 times.

Neurite outgrowth: HN initiation and cortical initiation assays

Human iPSC-derived glutamatergic-enriched cortical neurons (iCell GlutaNeurons; Fujifilm Cellular Dynamics) or rat primary cortical neurons prepared in house from postnatal day 0 Long Evans pups were plated in 96-well plates. Two hours after plating, cells were exposed to 10 concentrations of each test chemical. Each experiment used 3 plates in the same culture to provide 3 technical replicates. After 48 h of exposure, cells were fixed with 4% PFA and stained with Hoechst-33342 and β -III-tubulin (neurites) by ICC method. The number of cells (neurons) per field, total neurite length per neuron, number of neurites per neuron, and number of neurite branch points per neuron were determined by automated image acquisition and analysis using the Cellomics ArrayScan VTI HCS Reader. Each plate contained 3 concentrations of the endpoint specific control for neurite outgrowth (1, 3, 10, and 30 μ M Rac 1 inhibitor) and the vehicle control. For full details, see Druwe et al. (2015) and Harrill et al. (2018). All animal protocols were approved by the Office of Research and Development's Institutional Animal Use and Care Committee (IACUC) at the US EPA in Research Triangle Park, NC.

Neurite maturation and synaptogenesis: cortical maturation and synaptogenesis assays

Rat primary cortical neurons derived from postnatal day 0 Long Evans pups were plated on 96-well plates and allowed 2 h to attach prior to media exchange. After 5 d in vitro (DIV), cytosine arabinoside (1 μ M) was added to the wells to arrest glia cell growth. Between 7 to 12 DIV, cells were exposed to 9 concentrations of each test chemical in triplicate. On DIV 12, the cultures were fixed in 4% PFA and nuclei (Hoechst 33342), synapses (synaptophysin), and dendrites (MAP2) were stained using ICC method. Each experiment used 3 plates in the same culture to provide 3 technical replicates. A Cellomics ArrayScan VTI HCS Reader (Thermo Fisher Scientific) was used for automated image acquisition and analysis of cortical maturation and synaptogenesis. Measurements for neurite maturation included the number of neurites, neurite length, and the number of neurite branch points per neuron based on the morphology of synaptophysin- and MAP2-labeled neurons. Measurements for synaptogenesis included the number of presynaptic puncta in the cell body compartment and the neurite compartment, the total number of synapses, and the number of neurite-associated puncta per unit length of neurite. Cell viability was assessed as the number of cells per field at the time of fixation in comparison to the vehicle control. Endpoint-specific controls included a well containing the endpoint-selective control for impaired synaptogenesis (10 μ M sodium orthovanadate) and the vehicle control. For full details, see Harrill et al. (2011).

Neural network formation: multiwell microelectrode array network formation assay

Rat primary cortical neurons derived from PND 0 Long Evans pups were cultured on a 48-multiwell plate (Axion Biosystems Inc., Atlanta, GA). After 2 h, cells were exposed to 7 concentrations of each test chemical. Each experiment used triplicate replicates to yield 3 technical replicates. Over the course of 12 DIV, networks of electrically active excitatory and inhibitory neurons

form spontaneously and become progressively more organized as synaptic connections are formed (Brown et al. 2016). Each well contained 16 microelectrodes that allow for extracellular recordings of electrical activity in the cells plated on top of the electrodes. Fifteen-minute recordings of spontaneous network activity were recorded on DIV 5, 9, and 12 using the Axion Biosystems (Atlanta, GA) Maestro 768-channel amplifier, a Middle-Man data acquisition interface, and Axion Integrated Studio (AxIS) software v2.5. Seventeen parameters relating to general activity, bursting activity, coordinated network connectivity were determined by analysis of each recording on each DIV. After recording on DIV 12, cell viability was determined in each well using the CellTiter-Blue (Promega) and LDH (Promega) assays. For full details, see (Shafer 2019).

Data analysis and concentration-response modeling

Concentration-response modeling was performed using the ToxCast Pipeline (tcpl; v3.2.0) (Feshuk et al. 2023). Key steps to the tcpl workflow are conducted in a stepwise process with results at each step written to the invitrodb database to maintain transparency for data processing. The testing results for samples presented herein have been made publicly visible in the invitrodb v4.3 rendering all levels of data accessible via database download and the US EPA Comptox Chemicals Dashboard (<https://comptox.epa.gov/dashboard>). The methods and outcomes for each level of the tcpl workflow include the following: (i) applying relevant data formatting, labeling data types (e.g. identifying control wells, test article wells, etc.), (ii) removing poor quality wells flagged by the testing facility (e.g. based on technical quality control review) and defining baseline values (e.g. computing plate-wise median response based on control wells), (iii) normalizing response values per assay endpoint data type based on user-defined standardized tcpl methods(s), (iv) conducting bidirectional curve fitting to 9 possible parametric models, (v) defining efficacy cutoffs for each assay endpoint based on standardized tcpl methods and determining winning curve-fit models to derive activity hit call (a continuous value between 0 and 1), and (vi) identifying curve-fitting flags to provide additional warning information (e.g. noisy data, statistical overfitting, or putative false positives or negatives). Curve-fitting results in invitrodb are stored as level 5 data and include the hit call value and AC50 value (concentration at 50% maximal activity used to summarize the potency of bioactivity). The baseline variability per assay endpoint was calculated as the median absolute deviation (bmad) across all vehicle controls and the lowest 2 testing concentrations from any test article ever run in the assay to yield a robust representation of baseline for setting bioactivity cutoffs. Most of the assay endpoints evaluated herein were reported in units of percent relative to vehicle control; these response values used for concentration-response modeling were calculated using Equation (1):

$$\text{resp.pc} = (\text{cval} - \text{bval}) / (\text{bval}) \times 100$$

where resp.pc (a term defined by tcpl methods) is the assay's response measure reported as percent of positive control, cval is the corrected response value (from tcpl level 2 normalization methods applied per assay endpoint), and bval is the baseline value (where baseline is defined by the median response of control samples per testing plate, as defined by tcpl level 2 methods per assay endpoint).

To render raw assay readouts from plate readers and detection software amenable to tcpl concentration-response modeling and hit calling, preprocessing was conducted per assay type as needed. Briefly, for the multiwell microelectrode array (mwMEA) network formation assay (NFA) custom R scripts (Cotterill et al. 2016; Martin et al. 2024) were used to compute 17 endpoints on each recording day (DIV 5, 7, 9, 12) and an area under the curve (AUC) metric was calculated to represent changes in activity over time (per endpoint and well). These AUC values were input as raw data for tcpl level 1. Preprocessing of the EPA high-content imaging assays (neurite outgrowth assay, synaptogenesis, proliferation, and apoptosis assays) mainly consisted of reformatting data to comply with tcpl layout (Harrill et al. 2018). Data preprocessing for the IUF and UKN assays were also consistent with previous descriptions (Masjosthusmann et al. 2020). Namely, the IUF NPC1 assay entailed computing median response values per plate and subtracting background (BrdU) for the NPC1 proliferation and viability endpoints. For the NPC2-5 endpoints, the median of response values was calculated per plate and data were normalized to the percent of vehicle controls.

DNT-IVB data interpretation

After concentration-response modeling, outputs from invitrodb levels 5 and 6 were compiled. First, concurrent cytotoxicity readouts per assay technology were evaluated to confirm that cytotoxicity did not confound any of the assay endpoint data. Having confirmed that neither ASM nor AA elicited any overt cytotoxicity, the level 5 hit call values were reviewed to identify putatively active endpoints with a hit call ≥ 0.9 (Feshuk et al. 2023). Putatively active assay endpoints having 4 or more level 6 cautionary flags, a fit category of 36 (indicating AC50 is extrapolated as being less than or equal to the minimum concentration tested), or a hit call of -1 (indicating a nonbiologically relevant fitting direction) were considered low confidence and insufficiently robust for interpretation. For a list of tcpl level 6 cautionary flags, see File S3, Table S3. For the remaining active assay endpoint, the AC50 values were retrieved to serve as a metric representing the potency of bioactivity.

Results

In vivo DNT study findings

A summary of results from the in vivo DNT study with ASM is presented in File S1. Briefly, there were no deaths or adverse clinical observations for the dams fed ASM during gestation and lactation. Transient differences in maternal body weight (GD 11) and food consumption (PND 12-18) at 4,000 ppm (326.2 mg/kg bw/d) were noted. Reproductive parameters were unaffected by maternal treatment with ASM and there were no adverse clinical signs or adverse effects on litter size, pup survival, total litter weight, or pup body weight to PND 5. The body weight of offspring on PND 18, 22, and 29 was slightly lower (7% to 9%) in the 4,000-ppm group. The effect on body weight was seen to occur when pups were weaning; thus, it was considered to be resulting from relatively high exposure to ASM from their own direct feed consumption in addition to lactational exposure.

There were no effects of maternal ASM consumption on offspring across a multitude of endpoints including the time of vaginal opening or preputial separation, on the FOB, locomotor activity, or learning and memory. Furthermore, there was no effect on ASR amplitude on either PND 23 or 61 for F1 males, or for F1 females on PND 61, and no effect on startle response time to maximum amplitude in F1 animals on either time point.

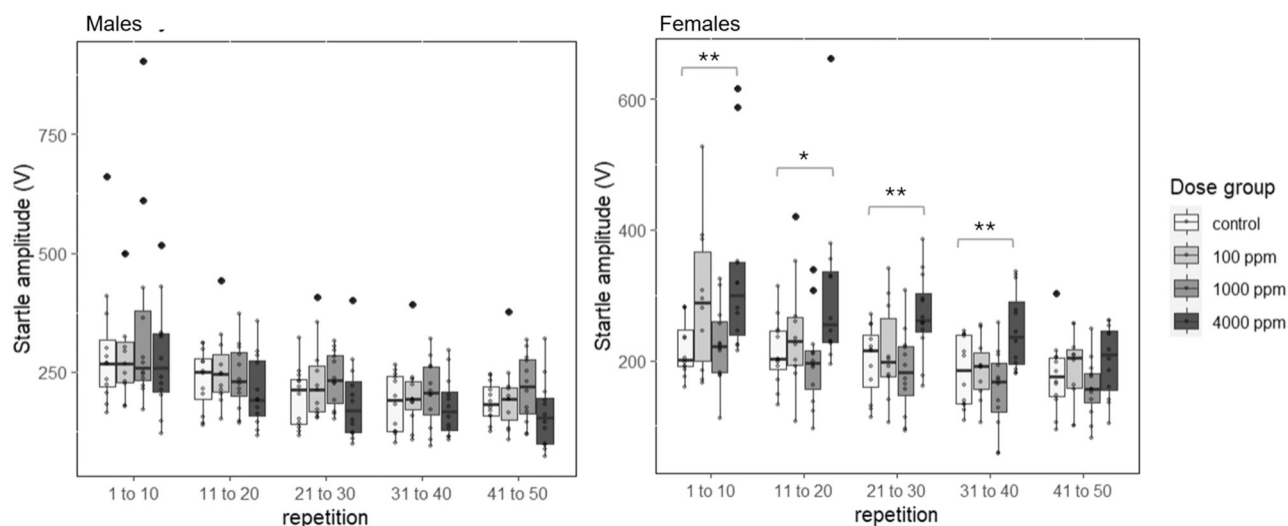


Fig. 2. Auditory startle response (ASR) in rats on postnatal day (PND) 23 following maternal treatment with ASM. Data include the mean startle response amplitude (V) across 5 repetitions for males (A) and females (B). One male or one female from each litter selected on PND 5 was assigned to auditory startle test ($N = 10$ to 12 pups/sex/group). The animals were assessed on PND 23 and 61 (data provided in [File S1](#)). An automated recording apparatus was used for the test. Boxplots represent 25th to 75th percentile of measurements, with the mean drawn in the box; the boxplot whiskers represent 5th and 95th percentiles; any measurements beyond the 5th or 95th percentile are plotted as points. Statistical analyses were carried out in SAS (1999); data were assessed by analysis of variance (ANOVA) using the MIXED procedure where * $P < 0.05$, ** $P < 0.01$ only for the highest dose group (4,000 ppm) versus vehicle control.

The test for ASR on PND 23 revealed a possible difference in performance of the F1 females in the 4,000-ppm group in comparison with the control group ([Fig. 2](#)). Plotting all individual measures in addition to the boxplots in [Fig. 2](#) illustrates a noticeable overlap of the range of values measured in the high dose group and the concurrent controls. Furthermore, there is uncertainty in the toxicological relevance of the increased ASR in females on PND 23 in isolation given the effect was not maintained at the later time point, and because there is evidence that the ASR may not be fully mature in the postweaning period ([PMRA 2021](#)).

There were no treatment-related microscopic findings in the nervous tissues of F1 animals. There were no effects on brain weights for F1 animals on either PND 12 or 63. There were no treatment-related effects on brain morphometry on PND 12. Reductions in the thickness of the dorsal cortex (level 3) and molecular layer of the cerebellum in both the prepyramidal fissure and preculminate fissure were identified in F1 male animals at PND 63, in the dorsal cortex at ≥ 100 ppm (8.2 mg/kg bw/d), in the prepyramidal fissure at $\geq 1,000$ ppm (82 mg/kg bw/d), and in the preculminate fissure at 4,000 ppm (326.2 mg/kg bw/d) ([Fig. 3](#)).

Overall, the *in vivo* DNT study was largely negative for the abundance of endpoints measured ([File S1](#)) with debate surrounding biological significance of the low-magnitude morphometric and startle effects. There is a high degree of uncertainty surrounding the *in vivo* findings highlighting not only why the OECD test guideline update was needed, but also the challenges in interpreting *in vivo* DNT endpoints generated preceding the change.

PBPK modeling

As the DNT-IVB comprised assays derived from both rat and human cells, appropriate testing concentrations for each species were established to reflect fetal exposure following maternal exposure. By carefully defining the assumption of 13 wk of gestation as a critical stage at which embryonic exposure can no longer be adequately represented by maternal blood concentrations

and instead requires separate fetal predictions ([Kapraun et al. 2022](#)), and leveraging a parallelogram approach to robustly build and evaluate PBPK models, this study demonstrates a unique context of use for such predictive modeling. The fit-for-purpose approach allows for ensuring that all *in vitro* test systems are evaluated at appropriate testing concentrations.

To predict the concentrations of ASM and AA in blood and brain, 2 rat dietary exposure scenarios were simulated using the rat PBPK model and 2 human dietary exposure scenarios for adult and fetus were simulated using the adult human and gestational PBPK models. For rats, the models included daily dietary oral consumption until reaching steady state at the low dose (100 ppm; 8.2 mg/kg bw/d) or highest dose level (4,000 ppm; 326.2 mg/kg bw/d). For humans, the models included the daily dietary oral consumption for 14 d or 40 gestational weeks at the levels of the low HED (1.51 mg/kg bw/d) or the highest HED (62.7 mg/kg bw/d; [Fig. 4](#)). For illustrative purposes, [Fig. 4](#) presents the model-predicted human fetal concentrations of ASM and AA at the highest HED, highlighting negligible fetal exposure to ASM and the temporal dynamics of AA, with decreasing concentrations in brain tissue and increasing concentrations in blood over gestational period. The model predicted that fetal exposure, in both systemic circulation and brain tissue, was generally lower compared with maternal exposure, indicating that maternal exposure may serve as a conservative surrogate for fetal exposure.

Overall, the modeled systemic exposure to parent compound ASM is significantly lower than AA due to rapid and extensive metabolism of ASM by carboxylesterase activity in peripheral tissue in both species ([Table 1](#)). This emphasized the need to evaluate AA as a test article *in vitro* when trying to investigate the effects of maternal ASM exposure. The daily average concentrations of ASM in blood and brain were predicted to be below $1 \mu\text{M}$ for all exposure scenarios. For metabolite AA, at the highest HED dose level, the estimated 95th percentile of maximum daily average human fetal blood and brain concentrations were predicted to be $\leq 100 \mu\text{M}$, and for rats $< 140 \mu\text{M}$ (blood) and $\leq 30 \mu\text{M}$ (brain).

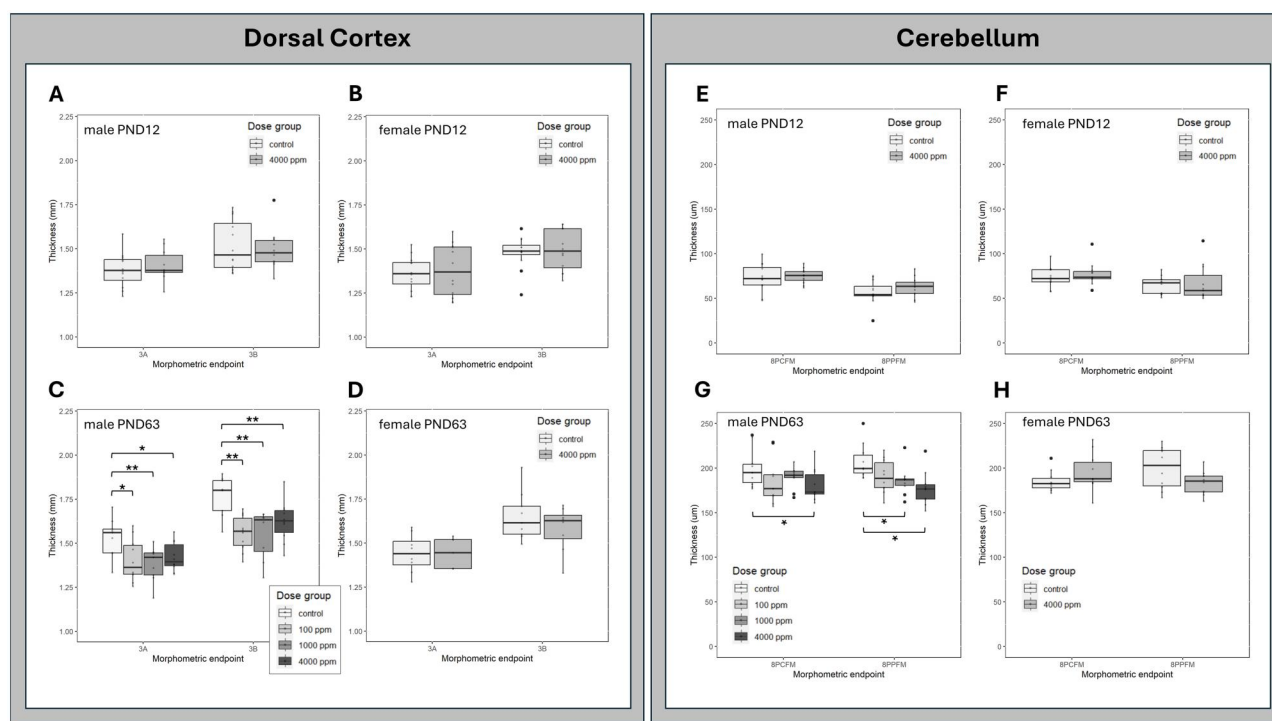


Fig. 3. Dorsal cortex and cerebellum morphometric measurements from ASM treatment in the in vivo DNT study. (A–D) Dorsal cortex morphometric endpoints include the thickness of the dorsal cortex on postnatal day (PND) 12 (Panels A and B) and PND 63 (Panels C and D) Dorsal cortex endpoint definitions: 3A, dorsal cortex at level 3 section of brain, inner origin at most dorsal point of external capsule parallel to the midsagittal line (average of left and right); 3B, dorsal cortex at level 3 section of brain, inner origin at a point cut by a line drawn at ~45 degrees to midsagittal line (average of left and right). (E–H) Cerebellum morphometric endpoints include the thickness of molecular layer of the preculminate fissure (8PCFM) and prepyramidal fissure (8PPFM) on PND 12 (Panels E and F) and PND 63 (Panels G and H). Cerebellum endpoint definitions: 8PCFM, cerebellum at level 8 section of brain, molecular layer of preculminate fissure; 8PPFM, cerebellum at level 8 section of brain, molecular layer of prepyramidal fissure. All measurements were conducted in male and female rats measured on PND 12 and 63 (N = 1 pup/sex/litter to give a total of 10 pups/sex/group). Boxplots represent 25th to 75th percentile of measurements, with the mean drawn in the box; the boxplot whiskers represent 5th and 95th percentiles; any measurements beyond the 5th or 95th percentile are plotted as points. Statistical analyses were carried out in SAS (1999); data were assessed by analysis of variance (ANOVA) where * $P \leq 0.05$, ** $P \leq 0.01$ versus vehicle control.

Thus, to ensure that half-log-spaced in vitro testing concentration series for evaluating ASM and AA in the DNT-IVB encompassed the predicted fetal exposures, a maximal testing concentration of 30 or 100 μM was used for ASM testing and 100 or 300 μM was set as the top testing concentration for AA in human- and rat-based assay systems, respectively (Table 1).

DNT-IVB results

Overall, most in vitro test systems in the DNT-IVB yielded no bioactive findings for both ASM and AA, which aligns with the overall negative findings from the in vivo DNT study. The parent compound ASM was negative across the battery of in vitro assays that comprise the DNT-IVB, up to the highest concentrations tested (30 or 100 μM). The mammalian metabolite AA was negative in the battery up to 100 μM (for human cell-based test systems) and 300 μM (for rat-derived assays) except for the NFA assay. The NFA platform generates 17 assay endpoint readouts of which AA was active in 8 related to formation of neuronal network activity in rat primary cortical culture (Fig. 5). Bursting activity (AA active in 2/4 endpoints), general activity (1/4 endpoints), and network connectivity (5/9 endpoints) decreased in the absence of cytotoxicity following exposure for 12 DIV to AA. The active NFA hit calls were almost exclusively driven by response in the highest tested concentration (300 μM), where a near-complete abolishment of activity was observed for all the active endpoints, thus, AC50s were quite high ranging from 96.1

to 150 μM . This AC50 range was comparable to the maximum systemic exposure level predicted in rat (118 μM) and was ~3.8- to 6.0-fold higher than the maximum daily brain exposure level predicted in rat (30 μM) from the highest dose group from the in vivo study. Notably, 6 of 8 active endpoints had tcpl level 6 cautionary flags (albeit fewer than 4 flags, and thus still considered active) relating to noisy data, borderline activity, or single-point hit that is only at the highest concentration tested.

The goal of conducting testing in the DNT-IVB was to provide insight into processes (KNDPs) that may underlie any putative DNT concern arising from ASM exposure. Increased ASR was observed only in females from the highest dose group on PND 23, with no effects noted in males at any dose or age, on PND 61 in females, or on startle response time to maximum amplitude across any time point, dose group, or sex. For this behavioral effect, relevant KNDPs in the DNT-IVB may include neurite maturation/synaptogenesis and network formation, both of which are critical for facilitating information flow necessary for modulating behavior. Though there may be other etiologies that may underlie ASR, all DNT-IVB testing for ASM and AA showed no activity in neurite maturation/synaptogenesis and the only active effects were found in some NFA endpoints elicited by AA. These active endpoints represent only a subset of relevant DNT-IVB endpoints for corroborating functional network activity. Active NFA endpoints were predominately driven by the highest concentration tested (300 μM), which exceeds the in vivo highest

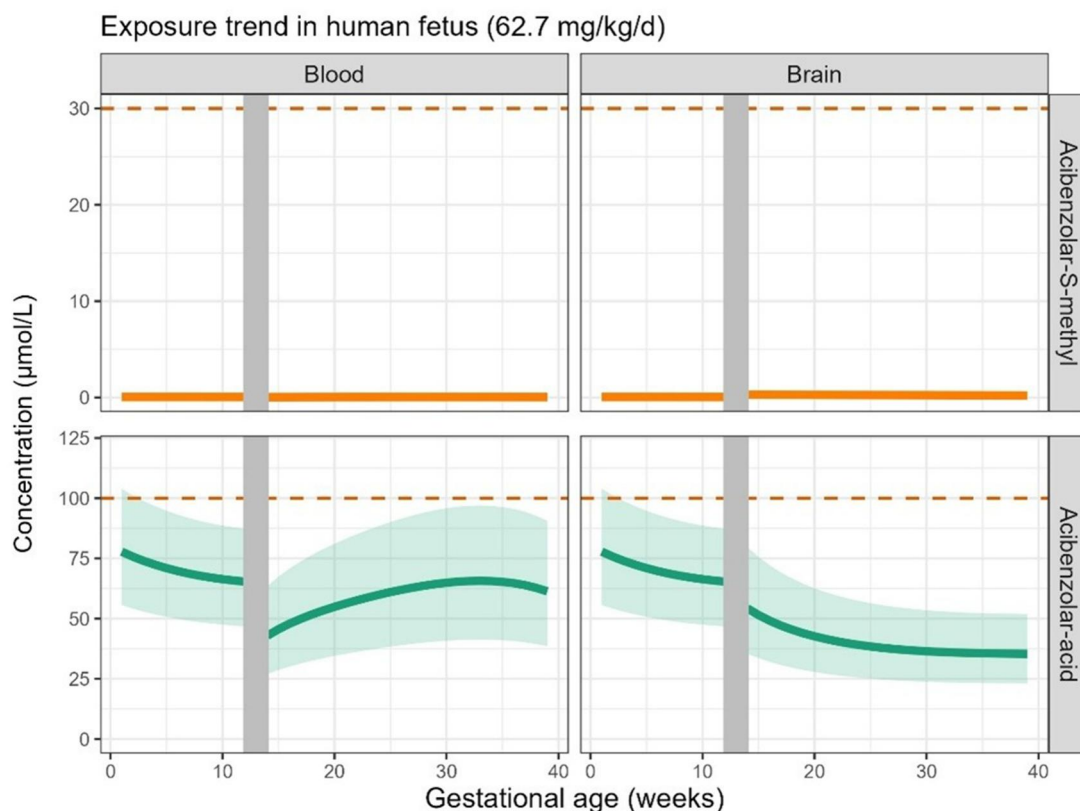


Fig. 4. Simulated weekly average concentration time course profiles of ASM and AA. Simulation of human fetal blood (systemic) and brain (tissue) concentrations for ASM and AA during gestational weeks 1 to 40 given maternal daily dietary exposure to 62.7 mg/kg bw/d ASM. The dashed horizontal line represented the upper testing limit of the in vitro DNT battery for human-based assay systems (e.g. 30 μM for ASM and 100 μM for AA). The grey vertical bar represents gestational week 13 where distinction is made for how fetal concentrations were predicted: Before 13 wk embryonic concentrations were approximated using the modeled maternal blood concentrations from the gestational PBPK model; after 13 wk, fetal blood and brain concentrations were derived from the fetal submodel within the gestational PBPK model. The solid line represents the median modeled concentrations of ASM (orange) or AA (teal); the teal shading represents 95% CI.

dose (4,000 ppm, modeled to be equivalent to 30 and 100 μM in rat and human brain, respectively), and therefore, is not biologically relevant to inform on the increased ASR in F1 females.

The in vivo DNT study with ASM also identified morphometric findings in males suggesting slight thinning of the dorsal cortex and molecular layers of the cerebellum by PND 63, but not PND 12. Therefore, KNDPs informing on processes extending into the postnatal period would likely be most relevant to contextualize this finding. These KNDPs include neurogenesis, gliogenesis, myelination, neuronal arborization, synaptogenesis, and apoptosis, which are evaluated in assays focused on proliferation (hNP1 and NPC1 assays), differentiation (NPC2-5 and UKN2 assays), neurite outgrowth (NPC4 and UKN4-5 assays), and neurite maturation/synaptogenesis (Cortical Maturation and Synaptogenesis assay). The in vitro testing results for these assays were completely negative for both ASM and AA. Effects on linear morphometry were observed in F1 males at doses as low as 100 ppm; at this dose, the 95th percentile of maximum daily average concentration of AA in rat and human fetal brain was modeled to be $\leq 2.5 \mu\text{M}$. The DNT-IVB showed no true evidence of DNT response at concentrations where the morphometric effects were observed.

Considering that the AC50s obtained in the NFA assay with AA were >44-fold higher than the modeled brain concentrations at the in vivo low dose (where equivocal brain morphometry effects were observed), the overall weight of evidence, including the confounding factor of variable tissue storage in fixative, there

is a low level of concern that the cortical and cerebellar thinning observed in the ASM in vivo study was elicited via any KNDP modeled in the DNT-IVB.

Discussion

In the present case study, the DNT-IVB was used to assess the biological plausibility that compound exposure rather than technical artifact led to the in vivo DNT study findings. With this goal, anchoring our investigation to in vivo dosimetry was a key study design consideration. By employing PBPK modeling and known metabolism information, testing of both the parent compound (ASM) and its primary mammalian metabolite (AA) included concentrations that were relevant to the in vivo DNT study. However, it is also important to note real-world exposures compared with the aforementioned in vivo DNT study. Exposure to ASM may be occupational (e.g. agricultural workers), via food, water, residential, or indirect exposure (e.g. spray drift). Based on the US EPA assessment during the re-registration of ASM, the highest exposure for workers is 0.041 mg/kg/d and for nonoccupational exposures, 0.028 mg/kg/d (US EPA 2017, 2018). This would lead to in vivo concentrations far less than concentrations associated with any effects in the in vivo DNT study or the DNT-IVB assays.

The primary challenge with interpreting the original in vivo brain morphometry findings stems from the stepwise tissue processing approach used wherein low- and mid-dose group

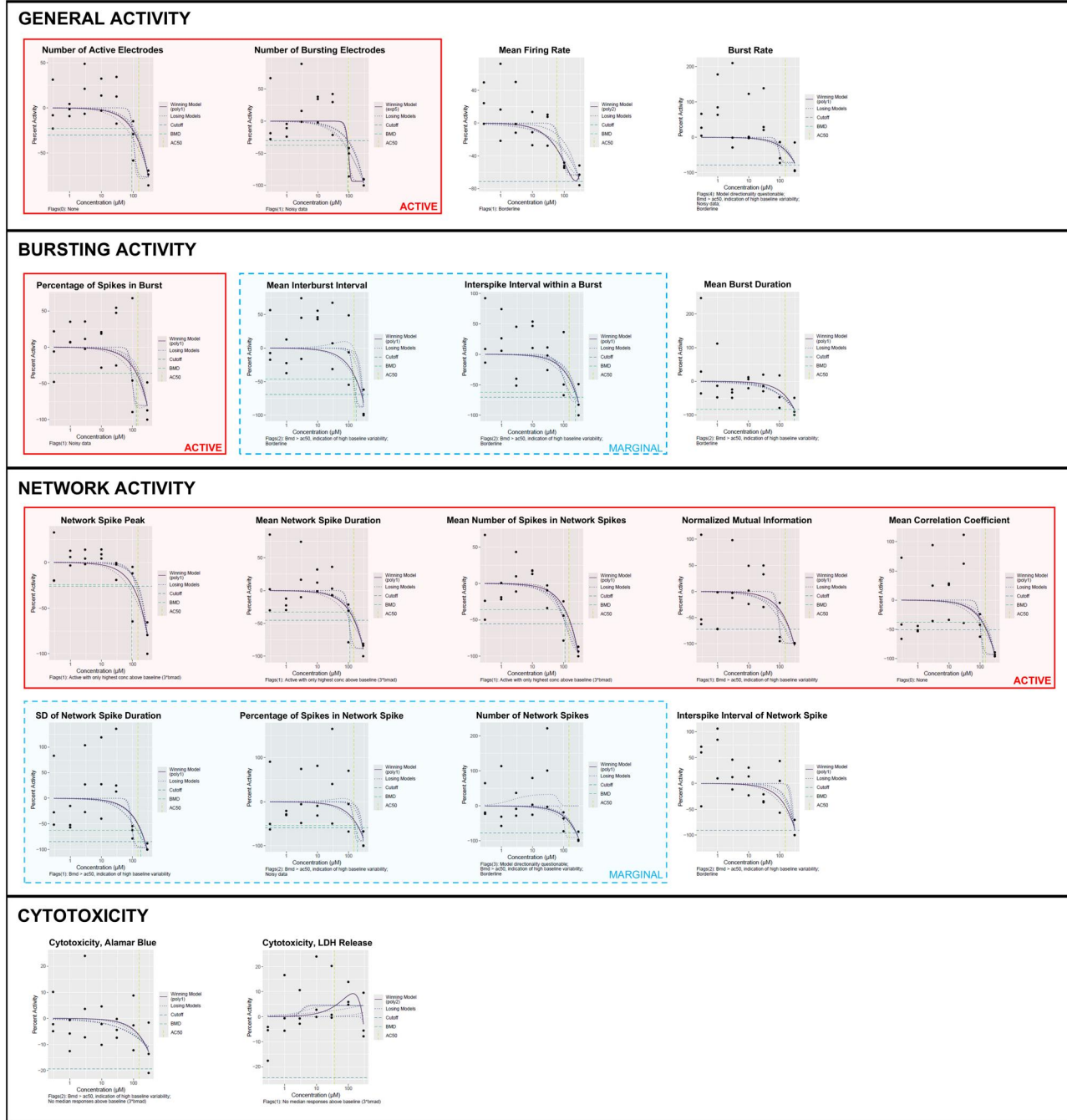


Fig. 5. Concentration-response curves for AA-elicited inhibition of electrical activity as measured by mwMEA NFA recordings in rat primary cortical culture. Endpoints are grouped by text headings to organize endpoint types. Red boxes surround active endpoints (hit call ≥ 0.9), dashed blue box encompass marginal endpoints (hit call 0.8 to 0.9), and no box surrounds concentration-response curves of inactive endpoints (hit call < 0.8). Dotted horizontal lines represent assay endpoint-specific cutoffs which response must exceed for active hit calling, evaluating how much responses exceed this threshold is an indication of effect efficacy. Empirical endpoints include mean interburst interval, interspike interval within a burst, mean burst duration, network spike peak, mean network spike duration, mean number of spikes in network spikes, and interspike interval of network spike; other displayed parameters are derived from this empirical data. Data points represent mean $\pm$ SEM ($N = 3$) and plots are adjusted for visualization of response detected (note varying y-axis ranges). Associated flags denote potential data considerations: Flag 5 (questionable model directionality), flag 9 (high baseline variability), flag 10 (noisy data), flag 11 (borderline activity).

morphometric findings were potentially artifactual resulting from prolonged storage in fixative. Morphometric measurements are highly susceptible to fixation artifacts and other experimental factors which, if not standardized, can lead to misinterpretations of potential DNT effects (Garman et al. 2016). Studies have shown that prolonged storage in fixative leads to significant tissue shrinkage, with varying effects across different brain regions.

Outer brain regions (e.g. cortex) experience greater shrinkage compared with inner regions, likely due to differences in myelin and water content (Quester and Schröder 1997). Furthermore, murine brain tissue fixed in formalin exhibited brain region-dependent variances up to $\sim 30\%$ (Wehr et al. 2015), and extended fixation durations (5d vs 24h) resulted in significant decreases in total brain volume, primarily affecting grey matter

(de Guzman et al. 2016). It is well-established that processing tissues sequentially (e.g. controls and high dose first, lower doses later) introduces fixation artifacts that compromise the validity of comparisons between groups (Bolon et al. 2006; Tsuji and Crofton 2012; Garman et al. 2016; Li et al. 2017). This known methodological limitation provided a strong rationale for updating the OECD Test Guideline for the conduct of the DNT study (OECD 2007). Furthermore, for studies conducted prior to study refinement and potentially confounded for interpretation of these morphometric endpoints, seeking alternative evidence such as the DNT-IVB rather than repeating studies can offer insight on the biological plausibility of the observed morphometric changes. In this regard we demonstrate an application of the DNT-IVB focused on interpreting relevant KNDPs as they relate to in vivo apical findings rather than utilizing the DNT-IVB for any screening or adverse outcome pathway (AOP)-based interpretation as has been conducted within different contexts of use (Blum et al. 2023; EFSA et al. 2024).

Although the entire DNT-IVB with 17 assay test systems was conducted to ensure a comprehensive evaluation, we focused our interpretation of in vitro results on assays evaluating KNDPs relevant to observed effects in vivo. The in vivo finding of reduced cortical and cerebellar layer thickness on PND 63, with limited findings at PND 12, points toward potential disruption of postnatal brain development in the immature pup. Neurogenesis, gliogenesis, myelination, synaptogenesis (including synaptic pruning), and apoptosis continue after birth and are KNDPs covered by the DNT-IVB. These processes must also take place for network formation and function to be normal in mammals, which is modeled by the NFA assay (Kaufmann 2000; Andersen 2003; van Dyck and Morrow 2017; Zeiss 2021). These KNDPs occur in overlapping temporal windows and are responsible for the more than sixfold increase in brain mass from birth to adulthood in rats (Bandeira et al. 2009). Although many of these processes initiate prenatally, their progression and refinement extend well into the postnatal period. This includes ongoing neurogenesis in specific regions (e.g. hippocampus, cerebellum), extensive gliogenesis and myelination (peaking around PND 20–30), and the dynamic processes of synaptogenesis, neuronal arborization, and synaptic pruning which extend into adolescence/adulthood in rodents (Bayer 1985; De Fepipe et al. 1997; White and Barone 2001; Dimou et al. 2008; Faust et al. 2021). Perturbations in these processes can impact brain morphology and function in rodents (Griffiths et al. 1998; Mathis et al. 2003; Roy et al. 2005; Corrales et al. 2006; Wiegrefe et al. 2022) and are associated with developmental disorders in humans (Kaufmann 2000; Dindot et al. 2008; Schneider et al. 2024).

The DNT-IVB includes endpoints evaluating relevant fundamental cellular mechanisms that drive brain development during both prenatal and postnatal periods, making it a valuable tool for studying how neurodevelopmental processes might influence brain morphometry. For instance, neurogenesis, including its postnatal activity in restricted regions such as the hippocampus and cerebellum, is assessed using the NPC1 and hNP1 proliferation assays modeling neural progenitor proliferation, and the NPC3 assay modeling differentiation. Neurite outgrowth and arborization, critical components of synaptogenesis extending postnatally, were assessed in 5 different assays within the DNT-IVB (NPC4, UKN4, UKN5, hN Initiation, Cortical Initiation), evaluating parameters like neurite count, length, and branching. Synaptogenesis is assessed in the Cortical Maturation and Synaptogenesis assay. Although dedicated myelination assays are not included, the battery includes oligodendrocyte

differentiation (NPC5) and radial glia migration assays (NPC2a), relevant to gliogenesis. Furthermore, several assays (e.g. Cortical Initiation, Cortical Maturation, NFA) use mixed cultures containing glia, meaning observed neuronal effects could potentially be influenced by glial responses (Mauch et al. 2001; Suzuki et al. 2007; Fester et al. 2009). Finally, apoptosis is effectively modeled in the hNP1 apoptosis and proliferation assay, with additional apoptotic markers embedded in cytotoxicity endpoints across other DNT-IVB assays.

Comparing the in vivo and in vitro findings, the absence of major morphometric alterations at PND 12 in vivo suggested that prenatal KNDPs like proliferation and migration were likely unaffected, consistent with the negative findings for both ASM and AA in the in vitro DNT-IVB proliferation and migration assays. The slight thinning of cortical and cerebellar layers observed only at PND 63 in vivo initially raised concern about potential disruption of postnatal KNDPs. However, the DNT-IVB results showed no effects of either ASM or AA on the broad suite of assays relevant to these processes that extend postnatally—including neurogenesis, differentiation, neurite outgrowth/arborization, synaptogenesis, apoptosis, and oligodendrocyte differentiation. These processes must also take place for proper network formation and function to occur, which was modeled in the NFA assay. Although activity in this assay was observed for the metabolite AA, the concentration where activity was observed (300 μ M) is above the systemic and brain exposure level predicted in rats at the exposure range where effects on brain morphometry were observed (i.e. at the low and high in vivo dose, the modeled rat brain concentrations were ≤ 1 and ≤ 30 μ M, respectively). Collectively, the overall WoE does not support the hypothesis of a treatment-related effect on brain morphometry in F1 males on PND 63 at any dose level. Considering the known potential for fixation artifacts in the antiquated in vivo study methodology, these artifacts remain the most plausible explanation for the observed morphometric differences between dose groups.

We also considered the potential relationship between the low potency effects of AA in the NFA assay (AC50s 96 to 150 μ M) and the transient increase in ASR amplitude seen in the high-dose females in vivo. On PND 23, these females exhibited the expected decrease in startle amplitude over successive intervals, yet their responses remained significantly higher than controls at each time point, suggesting the animals adapted to the auditory stimulus but remained in a hyperactive state. The ASR is an unconscious reflex mediated by a circuit in which cochlear root neurons project to the caudal pontine reticular nucleus, which then activates spinal motor neurons to produce the response (Zheng and Schmid 2023). ASR amplitude can be increased by neuronal hyperexcitability stemming from impaired inhibitory signaling (e.g. GABAergic or glycinergic) or altered voltage-gated ion channel kinetics. Chemicals acting on these early key events are typically detected in the NFA assay (Shafer 2019; Carstens et al. 2022; Martin et al. 2024) demonstrating that these modes of action are represented within the DNT-IVB. It is of note that the NFA measures spontaneous neuroactivity rather than stimulus-evoked activity involving auditory and motor circuits; thus, it remains unclear how reduced spontaneous activity in cortical cultures predicts a hyperactive stimulus-evoked reflex. Although the NFA is designed to capture a broad range of targets and pathways, it may not cover every mode of action contributing to the ASR. Confidence in using the DNT-IVB to contextualize ambiguous in vivo findings relies on a WoE approach that acknowledges assay limitations and considers the alignment of all available

Table 1. Model-predicted 95th percentile of maximum daily average internal concentrations in rat and human fetal blood and brain, and selected testing concentration ranges in DNT-IVB assays.

Compound	Species	Maximum modeled daily average in vivo concentrations (μM)		Maximum testing concentration (μM) in DNT-IVB assays
Acibenzolar-S-methyl (ASM)	Rat ^a	Blood	0.02	30 ^b
		Brain	0.12	
	Human fetus	Blood	0.07	
		Brain ^c	0.07	
Acibenzolar acid (AA)	Rat ^a	Blood	139	300
		Brain	30	
	Human fetus	Blood	100	100
		Brain ^c	100	

^a Blood and brain values of ASM and AA in rat are based on the nonpregnant adult rat PBPK values.

^b For UKN assays (UKN2, UKN4, UKN5), the top testing concentration was 100 μM AA.

^c Before 13 wk, embryonic concentrations were approximated using the modeled maternal blood concentrations from the gestational PBPK model.

data with relevant KNDPs modeled in the battery. The ASR effect was considered ambiguous due to its transient nature and its restriction to the female sex. Activity in the NFA assay was limited to a very high concentration (300 μM) and lacked corroboration from other assays in the battery. Specifically, no effects were observed in assays modeling upstream KNDPs essential for proper network formation and function. Therefore, while we cannot exclude a mechanistic link, the WoE indicates that the observed high-concentration NFA findings hold limited significance for understanding the transient ASR result in light of all lines of evidence considered.

Lastly, PBPK modeling indicated that the 95th percentile of maximum 24 h average AA concentrations predicted in rat and human brain (30 and 100 μM , respectively) were below or close to the in vitro bioactive concentration range (effects at only top testing concentration, 300 μM and AC50s ranging 96 to 150 μM) obtained in the NFA. Our straightforward approach to the PBPK modeling herein provides an estimate of the nominal concentrations expected. However, it should be acknowledged that there are ongoing efforts to refine such modeling, especially to support in vitro to in vivo extrapolations for interpreting the DNT-IVB specifically (Paini et al. 2024). Although several in vitro disposition models have been developed to describe mechanistic process within in vitro assays (Armitage et al. 2014; Honda et al. 2019; Proença et al. 2019), we did not apply them mainly due to insufficient assay-specific information (e.g. dynamic changes in cell volume, lipid content, cell differentiation, degradation, and diffusion) which can significantly influence chemical distribution and partitioning behavior over time resulting in substantial uncertainty in chemical distribution modeling. Given these limitations, our use of nominal concentrations remains a valid and informative approach for interpreting in vitro bioactive concentration estimates. The reported average dietary dose levels (mg/kg/d) from the in vivo DNT study were used as model inputs, and temporal changes in feed intake during gestation and lactation were not explicitly represented; the resulting internal dose estimates should therefore be interpreted as reflecting the average administered dose levels rather than day-specific changes in exposure over the course of the study. Future work may inform refinement to the application of PBPK for DNT-IVB applications specifically and comparisons of custom models with established commercially available PBPK platforms would also be informative.

Interpreting DNT-IVB data relies on a WoE approach (OECD 2023), where isolated, low-potency findings lacking corroboration carry less weight. In this context, the AA NFA effect occurred without any corresponding effects on the upstream cellular

processes foundational for establishing functional neural networks. Furthermore, these in vitro network effects were observed at concentrations considerably higher than the PBPK-modeled fetal brain concentrations. Although we cannot exclude a mechanistic link, the WoE indicates that the observed high-concentration NFA finding holds limited significance for understanding the transient ASR result.

This case study demonstrates the potential fit-for-purpose application of the DNT-IVB, addressing uncertainties from a legacy in vivo study to refine hazard characterization and showcasing how these in vitro methods complement and enhance the understanding of in vivo DNT findings in DNT assessments. In this case, the DNT-IVB revealed no in vitro effects of ASM or AA relevant to the in vivo morphometry or ASR findings within relevant exposures. Applying the DNT-IVB informed by PBPK modeling can successfully contribute to a WoE to further contextualize spurious findings from in vivo studies.

Acknowledgments

The authors greatly appreciate the hard work and dedication of Marcel Leist at the University of Konstanz and for his conduct of the UKN assays, and Ms Theresa M. Freudenrich and Kathleen Wallace at the US EPA for their assistance in conducting the EPA assays.

Supplementary material

Supplementary material is available at *Toxicological Sciences* online.

Funding

This research was financially supported by Syngenta Crop Protection, LLC. This work was conducted as part of a Cooperative Research and Development Agreement between the US EPA Office of Research and Development and Axion Biosystems (CRADA 1554-24).

Conflicts of interest. The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Syngenta Group produces acibenzolar-S-methyl. Syngenta Crop Protection is a business unit of the Syngenta Group that funded the in vitro assessments and subsequent analyses. BLJ, YW, KA, CS, AHH, LMM, and ALK were current employees of Syngenta Crop Protection for the

duration of this work. EF is a shareholder of the company DNTOX which provides DNT IVB assay services.

Disclaimer

The United States Environmental Protection Agency (U.S. EPA) has subjected this article to Agency administrative review and approved it for publication. Mention of trade names or commercial products does not constitute endorsement for use. The views expressed in this article are those of the authors and do not necessarily represent the views or policies of the US EPA.

References

- Andersen SL. 2003. Trajectories of brain development: point of vulnerability or window of opportunity? *Neurosci Biobehav Rev.* 27:3–18.
- Armitage JM, Wania F, Arnot JA. 2014. Application of mass balance models and the chemical activity concept to facilitate the use of in vitro toxicity data for risk assessment. *Environ Sci Technol.* 48:9770–9779.
- Bandeira F, Lent R, Herculano-Houzel S. 2009. Changing numbers of neuronal and non-neuronal cells underlie postnatal brain growth in the rat. *Proc Natl Acad Sci USA.* 106:14108–14113.
- Baumann J, Dach K, Barenys M, Giersiefer S, Goniwiecha J, Lein PJ, Fritsche E. 2015. Application of the neurosphere assay for DNT hazard assessment: challenges and limitations. In: Kang, YJ, editor. *Methods in pharmacology and toxicology*. Totowa, New Jersey, USA: Humana Press Inc. p. 1–29.
- Bayer S. 1985. Neuron production in the hippocampus and olfactory bulb of the adult rat brain: addition or replacement? *Ann N Y Acad Sci.* 457:163–172.
- Blum J, Masjosthusmann S, Bartmann K, Bendt F, Dolde X, Dönmez A, Förster N, Holzer A-K, Hübenthal U, Keßel HE, et al. 2023. Establishment of a human cell-based in vitro battery to assess developmental neurotoxicity hazard of chemicals. *Chemosphere.* 311:137035.
- Bolon B, Garman R, Jensen K, Krinke G, Stuart B. 2006. A 'best practices' approach to neuropathologic assessment in developmental neurotoxicity testing—for today. *Toxicol Pathol.* 34:296–313.
- Brown JP, Hall D, Frank CL, Wallace K, Mundy WR, Shafer TJ. 2016. Evaluation of a microelectrode array-based assay for neural network ontogeny using training set chemicals. *Toxicol Sci.* 154:126–139.
- Carstens KE, Carpenter AF, Martin MM, Harrill JA, Shafer TJ, Paul Friedman K. 2022. Integrating data from in vitro new approach methodologies for developmental neurotoxicity. *Toxicol Sci.* 187:62–79.
- Corrales CE, Pan L, Li H, Liberman MC, Heller S, Edge ASB. 2006. Engraftment and differentiation of embryonic stem cell-derived neural progenitor cells in the cochlear nerve trunk: growth of processes into the organ of corti. *J Neurobiol.* 66:1489–1500.
- Cotterill E, Hall D, Wallace K, Mundy WR, Eglén SJ, Shafer TJ. 2016. Characterization of early cortical neural network development in multiwell microelectrode array plates. *J Biomol Screen.* 21: 510–519.
- De Fepipe J, Marco P, Faorem A, Jones EG. 1997. Inhibitory synaptogenesis in mouse somatosensory cortex. *Cerebral Cortex.* 7:619–634.
- de Guzman AE, Wong MD, Gleave JA, Nieman BJ. 2016. Variations in post-perfusion immersion fixation and storage alter MRI measurements of mouse brain morphometry. *Neuroimage.* 142:687–695.
- Dimou L, Simon C, Kirchhoff F, Takebayashi H, Götz M. 2008. Progeny of Olig2-expressing progenitors in the gray and white matter of the adult mouse cerebral cortex. *J Neurosci.* 28:10434–10442.
- Dindot SV, Antalffy BA, Bhattacharjee MB, Beaudet AL. 2008. The angelman syndrome ubiquitin ligase localizes to the synapse and nucleus, and maternal deficiency results in abnormal dendritic spine morphology. *Hum Mol Genet.* 17:111–118.
- Druwe I, Freudenrich TM, Wallace K, Shafer TJ, Mundy WR. 2015. Sensitivity of neuroprogenitor cells to chemical-induced apoptosis using a multiplexed assay suitable for high-throughput screening. *Toxicology.* 333:14–24.
- EFSA. 2014. Conclusion on the peer review of the pesticide risk assessment of the active substance acibenzolar-S-methyl. *EFSA Journal.* 12:3691.
- EFSA; Coja T, Adriaanse P, Choi J, Finizio A, Giraudo M, Kuhl T, McVey E, Metruccio F, Paparella M. 2025. Scientific opinion on the application of physiologically based kinetic (PBK) modelling for the quantitative in vitro to in vivo extrapolation (QIVIVE) of developmental neurotoxicity in vitro battery (DNT IVB) data for pesticide active substances. *EFSA Journal.* 23:e9814.
- EFSA; Crofton KM, Paparella M, Price A, Mangas I, Martino L, Terron A, Hernández-Jerez A. 2024. A developmental neurotoxicity adverse outcome pathway (DNT-AOP) with voltage gate sodium channel (VGSC) inhibition as a molecular initiating event (MiE). *EFSA Journal.* 22:e8954.
- Faust TE, Gunner G, Schafer DP. 2021. Mechanisms governing activity-dependent synaptic pruning in the developing mammalian CNS. *Nat Rev Neurosci.* 22:657–673.
- Feshuk M, Kolaczowski L, Dunham K, Davidson-Fritz SE, Carstens KE, Brown J, Judson RS, Paul Friedman K. 2023. The ToxCast pipeline: updates to curve-fitting approaches and database structure. *Front Toxicol.* 5:1275980.
- Fester L, Zhou L, Bütow A, Huber C, Von Lossow R, Prange-Kiel J, Jarry H, Rune GM. 2009. Cholesterol-promoted synaptogenesis requires the conversion of cholesterol to estradiol in the hippocampus. *Hippocampus.* 19:692–705.
- Förster N, Butke J, Keßel HE, Bendt F, Pahl M, Li L, Fan X, Leung P-c, Klose J, Masjosthusmann S, et al. 2022. Reliable identification and quantification of neural cells in microscopic images of neurospheres. *Cytometry A.* 101:411–422.
- Garman RH, Li AA, Kaufmann W, Auer RN, Bolon B. 2016. Recommended methods for brain processing and quantitative analysis in rodent developmental neurotoxicity studies. *Toxicol Pathol.* 44:14–42.
- Griffiths I, Klugmann M, Anderson T, Yool D, Thomson C, Schwab MH, Schneider A, Zimmermann F, McCulloch M, Nadon N, et al. 1998. Axonal swellings and degeneration in mice lacking the major proteolipid of myelin. *Science.* 280:1610–1613.
- Gutbier S, May P, Berthelot S, Krishna A, Trefzer T, Behbehani M, Efremova L, Delp J, Gstraunthaler G, Waldmann T, et al. 2018. Major changes of cell function and toxicant sensitivity in cultured cells undergoing mild, quasi-natural genetic drift. *Arch Toxicol.* 92:3487–3503.
- Harrill JA, Freudenrich T, Wallace K, Ball K, Shafer TJ, Mundy WR. 2018. Testing for developmental neurotoxicity using a battery of in vitro assays for key cellular events in neurodevelopment. *Toxicol Appl Pharmacol.* 354:24–39.
- Harrill JA, Robinette BL, Mundy WR. 2011. Use of high content image analysis to detect chemical-induced changes in synaptogenesis in vitro. *Toxicol In Vitro.* 25:368–387.

- Hoelting L, Klima S, Karreman C, Grinberg M, Meisig J, Henry M, Rotshteyn T, Rahnenführer J, Blüthgen N, Sachinidis A, et al. 2016. Stem cell-derived immature human dorsal root ganglia neurons to identify peripheral neurotoxicants. *Stem Cells Transl Med*. 5:476–487.
- Holzer AK, Suci I, Karreman C, Goj T, Leist M. 2022. Specific attenuation of purinergic signaling during bortezomib-induced peripheral neuropathy In vitro. *Int J Mol Sci*. 23:3734.
- Honda GS, Pearce RG, Pham LL, Setzer RW, Wetmore BA, Sipes NS, Gilbert J, Franz B, Thomas RS, Wambaugh JF. 2019. Using the concordance of in vitro and in vivo data to evaluate extrapolation assumptions. *PLoS One*. 14:e0217564.
- Kapraun DF, Sfeir M, Pearce RG, Davidson-Fritz SE, Lumen A, Dallmann A, Judson RS, Wambaugh JF. 2022. Evaluation of a rapid, generic human gestational dose model. *Reprod Toxicol*. 113:172–188.
- Kaufmann W. 2000. Developmental neurotoxicity. In: Krinke G, editor. *The laboratory rat*. San Diego: Academic Press. p. 227–250.
- Koch K, Bartmann K, Hartmann J, Kapr J, Klose J, Kuchovská E, Pahl M, Schlüppmann K, Zühr E, Fritsche E. 2022. Scientific validation of human neurosphere assays for developmental neurotoxicity evaluation. *Front Toxicol*. 4:816370.
- Krebs A, van Vugt-Lussenburg BMA, Waldmann T, Albrecht W, Boei J, ter Braak B, Brajnik M, Braunbeck T, Brecklinghaus T, Busquet F, et al. 2020. The EU-ToxRisk method documentation, data processing and chemical testing pipeline for the regulatory use of new approach methods. *Arch Toxicol*. 94:2435–2461.
- Krug AK, Balmer NV, Matt F, Schönenberger F, Merhof D, Leist M. 2013. Evaluation of a human neurite growth assay as specific screen for developmental neurotoxicants. *Arch Toxicol*. 87:2215–2231.
- Kuepfer L, Fuellen G, Stahnke T. 2023. Quantitative systems pharmacology of the eye: tools and data for ocular QSP. *CPT Pharmacometrics Syst Pharmacol*. 12:288–299.
- Li AA, Sheets LP, Raffaele K, Moser V, Hofstra A, Hoberman A, Makris SL, Garman R, Bolon B, Kaufmann W, et al. 2017. Recommendations for harmonization of data collection and analysis of developmental neurotoxicity endpoints in regulatory guideline studies: proceedings of workshops presented at Society of Toxicology and joint Teratology Society and Neurobehavioral Teratology Society meetings. *Neurotoxicol Teratol*. 63:24–45.
- Lin H-C, Baltazar MT, Cable S, Ford LC, Middleton A, Nicol B, Punt A, Reynolds J, Rusyn I, Chiu WA. 2025. Sensitivity analysis of the inputs for bioactivity-exposure ratio calculations in a NAM-based systemic safety toolbox. *Nam J*. 1:100056.
- Lotharius J, Falsig J, Van Beek J, Payne S, Dringen R, Brundin P, Leist M. 2005. Progressive degeneration of human mesencephalic neuron-derived cells triggered by dopamine-dependent oxidative stress is dependent on the mixed-lineage kinase pathway. *J Neurosci*. 25:6329–6342.
- Martin MM, Carpenter AF, Shafer TJ, Friedman KP, Carstens KE. 2024. Chemical effects on neural network activity: comparison of acute versus network formation exposure in microelectrode array assays. *Toxicology*. 505:153842.
- Masjosthusmann S, Becker D, Petzuch B, Klose J, Siebert C, Deenen R, Barenys M, Baumann J, Dach K, Tigges J, et al. 2018. A transcriptome comparison of time-matched developing human, mouse and rat neural progenitor cells reveals human uniqueness. *Toxicol Appl Pharmacol*. 354:40–55.
- Masjosthusmann S, Blum J, Bartmann K, Dolde X, Holzer A, Stürzl L, Keßel EH, Förster N, Dönmez A, Klose J, et al. 2020. Establishment of an a priori protocol for the implementation and interpretation of an in-vitro testing battery for the assessment of developmental neurotoxicity. EFSA Supporting Publications. 17:EN-1938.
- Mathis C, Collin L, Borrelli E. 2003. Oligodendrocyte ablation impairs cerebellum development. *Development*. 130:4709–4718.
- Mauch DH, Nagler K, Schumacher S, Goritz C, Müller E-C, Otto A, Pfrieger FW. 2001. CNS synaptogenesis promoted by glia-derived cholesterol. *Science*. 294:1354–1357.
- Motwani HV, Frostne C, Törnqvist M. 2017. Parallelogram based approach for in vivo dose estimation of genotoxic metabolites in humans with relevance to reduction of animal experiments. *Sci Rep*. 7:17560.
- Nimtz L, Klose J, Masjosthusmann S, Barenys M, Fritsche E. 2019. The neurosphere assay as an in vitro method for developmental neurotoxicity (DNT) evaluation. In: Aschner M, Costa L, editors. *Cell culture techniques*. New York, NY, USA: Humana. p. 141–168.
- Nyffeler J, Karreman C, Leisner H, Kim YJ, Lee G, Waldmann T, Leist M. 2017. Design of a high-throughput human neural crest cell migration assay to indicate potential developmental toxicants. *ALTEX*. 34:75–94.
- OECD. 2007. Test No. 426: developmental neurotoxicity study. Paris, France: OECD Publishing.
- OECD. 2023. Initial recommendations on evaluation of data from the developmental neurotoxicity (DNT) in-vitro testing battery. Series on Testing and Assessment No. 377. Paris, France: OECD Publishing.
- Paini A, Lousse J, Sachana M, Tan C. 2024. S20-01 principles of quantitative in vitro to in vivo extrapolation (QIVIVE) applying physiologically based kinetic (PBK) modelling for DNT IVB. *Toxicol Lett*. 399:S37–S38.
- Pearce RG, Setzer RW, Strobe CL, Sipes NS, Wambaugh JF. 2017. <http://Rpackageforhigh-throughputtoxicokinetics>. *J Stat Softw*. 79:1–26.
- PMRA. 2011. Registration decision RD2011-07: acibenzolar-S-methyl. Ottawa, Ontario, Canada: Pest Management Regulatory Agency, Health Canada.
- PMRA. 2021. Proposed re-evaluation decision PRVD2021-08: tebuconazole and its associated end-use products. Ottawa, Ontario, Canada. Pest Management Regulatory Agency, Health Canada.
- Proença S, Paini A, Joossens E, Benito JVS, Berggren E, Worth A, Whelan M, Prieto P. 2019. Insights into in vitro biokinetics using virtual cell based assay simulations. *ALTEX*. 36:447–461.
- Quester R, Schröder R. 1997. The shrinkage of the human brain stem during formalin fixation and embedding in paraffin. *J Neurosci Methods*. 75:81–89.
- R Core Team. 2024. R: a language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing.
- Roy TS, Sharma V, Seidler FJ, Slotkin TA. 2005. Quantitative morphological assessment reveals neuronal and glial deficits in hippocampus after a brief subtoxic exposure to chlorpyrifos in neonatal rats. *Brain Res Dev Brain Res*. 155:71–80.
- Schmitt W. 2008. General approach for the calculation of tissue to plasma partition coefficients. *Toxicol In Vitro*. 22:457–467.
- Schmuck MR, Temme T, Dach K, de Boer D, Barenys M, Bendt F, Mosig A, Fritsche E. 2017. Omnisphero: a high-content image analysis (HCA) approach for phenotypic developmental neurotoxicity (DNT) screenings of organoid neurosphere cultures in vitro. *Arch Toxicol*. 91:2017–2028.
- Schneider R, Kogel AK, Ladopoulos T, Siems N, Krieger B, Bellenberg B, Gold R, Ayzenberg I, Lukas C. 2024. Cortical atrophy patterns in myelin oligodendrocyte glycoprotein antibody-associated disease. *Ann Clin Transl Neurol*. 11:2166–2175.

- Scholz D, Pörtl D, Genewsky A, Weng M, Waldmann T, Schildknecht S, Leist M. 2011. Rapid, complete and large-scale generation of post-mitotic neurons from the human LUHMES cell line. *J Neurochem*. 119:957–971.
- Shafer TJ. 2019. Application of microelectrode array approaches to neurotoxicity testing and screening. *Adv Neurobiol*. 22:275–297.
- Soetaert K, Petzoldt T, Setzer RW. 2010. Solving differential equations in R. *R J*. 2:5.
- Stiegler NV, Krug AK, Matt F, Leist M. 2011. Assessment of chemical-induced impairment of human neurite outgrowth by multiparametric live cell imaging in high-density cultures. *Toxicol Sci*. 121:73–87.
- Suzuki S, Kiyosue K, Hazama S, Ogura A, Kashihara M, Hara T, Koshimizu H, Kojima M. 2007. Brain-derived neurotrophic factor regulates cholesterol metabolism for synapse development. *J Neurosci*. 27:6417–6427.
- Szeto KX, Le Merdy M, Dupont B, Bolger MB, Lukacova V. 2021. PBPK modeling approach to predict the behavior of drugs cleared by kidney in pregnant subjects and fetus. *Aaps J*. 23:89.
- Tsuji R, Crofton KM. 2012. Developmental neurotoxicity guideline study: issues with methodology, evaluation and regulation. *Congenit Anom (Kyoto)*. 52:122–128.
- US EPA. 1998. Health effects test guidelines OPPTS 870.6300 developmental neurotoxicity study. Washington, DC, USA: United States Environmental Protection Agency.
- US EPA. 2006. Approaches for the application of physiologically based pharmacokinetic (PBPK) models and supporting data in risk assessment. Washington, DC, USA: United States Environmental Protection Agency.
- US EPA. 2007. Assessing approaches for the development of PBPK models of pyrethroid pesticides. Washington, DC, USA: United States Environmental Protection Agency.
- US EPA. 2017. Acibenzolar-S-methyl: draft human health risk assessment in support of registration review. Washington, DC, USA: United States Environmental Protection Agency.
- US EPA. 2018. Acibenzolar-S-methyl: revision to the draft human health risk assessment in support of registration review. Washington, DC, USA: United States Environmental Protection Agency.
- van Dyck LI, Morrow EM. 2017. Genetic control of postnatal human brain growth. *Curr Opin Neurol*. 30:114–124.
- Wehr HF, Bezrukov I, Wiehr S, Lehnhoff M, Fuchs K, Mannheim JG, Quintanilla-Martine L, Kohlhofer U, Kneilling M, Pichler BJ, et al. 2015. Assessment of murine brain tissue shrinkage caused by different histological fixatives using magnetic resonance and computed tomography imaging. *Histol Histopathol*. 30:601–613.
- White LD, Barone S. 2001. Qualitative and quantitative estimates of apoptosis from birth to senescence in the rat brain. *Cell Death Differ*. 8:345–356.
- WHO. 2010. Characterization and application of physiologically based pharmacokinetic models in risk assessment. Geneva, Switzerland: World Health Organization.
- Wiegrefe C, Wahl T, Joos NS, Bonnefont J, Liu P, Britsch S. 2022. Developmental cell death of cortical projection neurons is controlled by a Bcl11a/Bcl6-dependent pathway. *EMBO Rep*. 23:e54104.
- Zeiss CJ. 2021. Comparative milestones in rodent and human postnatal central nervous system development. *Toxicol Pathol*. 49:1368–1373.
- Zheng A, Schmid S. 2023. A review of the neural basis underlying the acoustic startle response with a focus on recent developments in mammals. *Neurosci Biobehav Rev*. 148:105129.